Beyond Gene Silencing: Developing an Engineered Ubiquitin-Proteasome System Enhancer (UPSE) for Targeted Degradation of Mutant Huntingtin as a Novel Approach to Combating Huntington's Disease

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Executive Summary

Huntington's disease (HD) is a devastating neurodegenerative disorder caused by a genetic mutation that leads to the production of a faulty protein called mutant huntingtin (mHTT). This abnormal protein accumulates in neurons, disrupting vital cellular processes and ultimately causing neuronal death. As the disease progresses, patients experience a range of debilitating symptoms, including involuntary movements, cognitive decline, and psychiatric disturbances. Currently, there is no cure for HD. However, recent advancements in gene therapy offer a promising approach to combat this disorder. One such study evaluated the efficacy of a gene therapy called AAV5-miHTT, which uses a modified virus to deliver a therapeutic molecule that reduces the levels of mHTT in the brain. While this approach showed promising results in animal models, there are limitations, such as the transient nature of the therapeutic effect and the potential for off-target effects. To overcome these limitations, I propose a novel treatment strategy that directly targets the mHTT protein for degradation. My approach involves engineering a specialized chimeric protein complex called the Ubiquitin-Proteasome System Enhancer (UPSE) with three different protein domains. The UPSE is designed to specifically recognize and bind to mHTT, marking it for destruction by the cell's natural protein degradation machinery. By selectively eliminating mHTT, the UPSE aims to alleviate the toxic effects of the mutant protein and slow disease progression. To deliver the UPSE to the brain, I propose utilizing functionalized exosomes—tiny vesicles naturally produced by cells. These exosomes will be engineered to display specific molecules that enable them to cross the blood-brain barrier and target the affected brain regions. Once inside the neurons, the UPSE can initiate the degradation of mHTT, potentially halting or reversing the disease process. By targeting the root cause of HD and employing a targeted delivery system, my approach has the potential to provide a more effective and long-lasting solution for patients suffering from this devastating disorder.

Biological Review

I. Physiological Background of Huntington’s Disease

Huntington's disease (HD) is a neurodegenerative disorder caused by a CAG trinucleotide repeat expansion in the huntingtin (HTT) gene, leading to the production of mutant huntingtin protein (mHTT) with an expanded polyglutamine tract (Bates et al., 2015). The accumulation of mHTT in neurons leads to various pathophysiological changes, including transcriptional dysregulation, mitochondrial dysfunction, and impaired synaptic transmission (Saudou & Humbert, 2016). One of the key pathways affected in HD is the brain-derived neurotrophic factor (BDNF) signaling pathway. mHTT disrupts the transport and secretion of BDNF, a crucial neurotrophic factor for the survival and function of striatal neurons as can be observed in Figure 1 with a comparison of intracellular and intercellular signaling pathways for a better understanding of HD disruptions (Zuccato et al., 2010). Additionally, mHTT impairs the function of the ubiquitin-proteasome system (UPS) and autophagy, leading to the accumulation of misfolded proteins and cellular toxicity (Ortega & Lucas, 2014). These pathophysiological changes ultimately result in the

selective degeneration of medium spiny neurons (MSN) in the striatum and cortical neurons which receive glutamatergic inputs from the cortex and thalamus, and play a key role in the direct and indirect pathways of the basal ganglia circuitry (Calabresi et al., 2014). Essentially, their loss results in the symptoms of HD which typically manifest in adulthood and progressively worsen over time. Patients with HD experience a triad of motor, cognitive, and psychiatric symptoms, which may include chorea (involuntary, jerky movements), rigidity, difficulty with speech and swallowing, memory deficits, executive dysfunction, depression, and apathy (Bates et al., 2015).

In healthy individuals, the balance between the direct and indirect pathways is essential for proper motor function. However, in HD, the accumulation of mHTT in MSNs leads to a series of physiological alterations that disrupt this balance. As the disease progresses, the chronic excitotoxicity caused by excessive glutamatergic signaling leads to the degeneration of MSNs, particularly those in the indirect pathway (Deng et al., 2014). The loss of MSNs in the indirect pathway results in disinhibition of the thalamus and increased thalamocortical feedback, which may underlie the characteristic chorea and other motor symptoms of HD (Plotkin & Surmeier, 2015). Furthermore, mHTT affects the energy metabolism of MSNs by impairing mitochondrial function and reducing ATP production (Damiano et al., 2010). This energy deficit renders MSNs more susceptible to oxidative stress and excitotoxicity, further contributing to their degeneration.

In summary, the physiological alterations in HD center around the dysfunction and degeneration of MSNs in the striatum, which leads to an imbalance in the direct and indirect pathways of the basal ganglia circuitry.

II. Pathological Background of Huntington's Disease

The underlying pathology of HD is centered around the aggregation and accumulation of misfolded proteins, particularly the mHTT. As a monogenic disorder, HD is caused by pathogenic variants that reduce the function or stability of a single protein by altering its three-dimensional structure.

The pathogenic variants in HD include point mutations, insertions, or deletions in the DNA sequence that encodes the protein, or changes in the non-coding DNA that interfere with gene splicing (Caron et al., 2018). These pathogenic variants result in the production of mHTT, which undergoes conformational changes due to misfolding and becomes toxic to neurons. The abnormal conformational transition from alpha helix to beta sheet exposes hydrophobic amino acid residues and promotes protein aggregation (Labbadia & Morimoto, 2013). This structure resembles the type of insoluble aggregates observed in the neurons of individuals with different neurodegenerative diseases. The accumulation of misfolded mHTT leads to the formation of intracellular inclusions and disrupts various cellular processes, including transcriptional regulation, mitochondrial function, proteasome activity, and synaptic transmission (Saudou & Humbert, 2016).

The aggregation and accumulation of misfolded mHTT ultimately leads to the selective degeneration of MSNs. The pathological mechanisms underlying the misfolding and aggregation of mHTT involve a complex interplay of cellular processes, including impaired protein degradation, oxidative stress, and neuroinflammation (Dayalu & Albin, 2015). These processes work in concert to create a toxic cellular environment that promotes the progressive loss of neurons and the onset of clinical symptoms. By understanding the intricate network of cellular dysfunction caused by mHTT, researchers aim to identify novel therapeutic targets and develop strategies to mitigate the deleterious effects of the mutant protein.

Figure 1. Intracellular and Intercellular Pathogenesis of HD. (A) Intracellular pathogenesis: mHTT forms aggregates, disrupting transcriptional regulation, protein degradation, and mitochondrial function. (B) Intercellular pathogenesis: mHTT in cortical neurons leads to excitotoxicity and oxidative stress, causing neuronal degeneration. Loss of cortical input and direct effects of mHTT on MSNs result in MSN dysfunction, disrupting basal ganglia circuitry. mHTT also impairs BDNF transport from cortical neurons to the striatum, increasing MSN vulnerability. Adapted from Ross & Tabrizi (2011).

III. Manipulation of Certain Cellular Interactions and Molecular Mechanisms

The complex pathogenesis of HD involves a multitude of cellular interactions and molecular mechanisms that contribute to the progressive degeneration of neurons. To develop effective therapies for HD, it is crucial to identify and manipulate key cellular and molecular targets that play a central role in the disease process; moreover, it is important to understand existing pathways that have been experimented with that relate to combating this monogenic disorder in addition to the underlying biology that comes with this. Comprehending these dynamics and their biological interconnections paves the way for the innovative strategy proposed later in this paper.

A. Targeting Specificities

To effectively combat HD, it is essential to target the root cause of the disease at the cellular and molecular level. The primary goal is to prevent the production of the mHTT protein, which contains an abnormally long polyglutamine (polyQ) tract that leads to its misfolding and aggregation that can be observed in Figure 2. This can be achieved by manipulating the genetic material responsible for the production of mHTT, specifically the expanded CAG repeat in the coding region of the HTT gene. By directly targeting the DNA, either through gene editing or gene silencing techniques, it is possible to reduce or eliminate the expression of the mutant gene. Gene editing often involves inducing double-strand breaks (DSBs) in DNA to facilitate targeted modifications, a process where the cellular repair machinery can introduce specific genetic changes. Alternatively, recent innovations like CRISPR-based 'base editors' achieve single base conversions without creating DSBs, reducing the risk of unintended mutations while still allowing precise genomic edits (Adli, 2018). Additionally, targeting the mHTT mRNA through RNA interference (RNAi) or antisense oligonucleotides (ASOs) can prevent the translation of the mutant protein, and these will be discussed more in the following section. Furthermore, strategies aimed at promoting the clearance of mHTT aggregates, such as enhancing the activity of the UPS or autophagy pathways, can help to mitigate the toxic effects of the mutant protein at the cellular level.

Figure 2. Structure of the HTT Protein and the Location of the Polyglutamine (polyQ) Tract. The HTT protein contains several HEAT (Huntingtin, Elongation factor 3, protein phosphatase 2A, and TOR1) repeats, which are involved in protein-protein interactions. The polyQ tract, encoded by the CAG trinucleotide repeat, is located near the N-terminus of the protein. The expansion of the polyQ tract beyond a critical threshold (>35 repeats) leads to the pathogenesis of HD. The HTT protein undergoes various post-translational modifications, such as proteolytic cleavage by caspases and calpains, ubiquitination, and phosphorylation, which can modulate its function and stability. Adapted from Ross & Tabrizi (2011).

Moreover, the extracellular matrix (ECM) also plays a role in the pathogenesis of HD and is definitely a potential target. In the context of HD, the ECM has been shown to undergo significant alterations, which may contribute to the progression of the disease. Studies have demonstrated that the expression of several ECM components, such as collagen, laminin, and chondroitin sulfate proteoglycans (CSPGs), is dysregulated in the brains of HD patients and animal models (Ritch et al., 2012). These changes in ECM composition can affect the survival, function, and plasticity of neurons, as well as the activity of glial cells. For example, the accumulation of CSPGs in the HD brain has been associated with impaired synaptic plasticity and reduced neuronal regeneration (Miguez et al., 2015). Targeting the ECM in HD could involve strategies aimed at modulating the expression or activity of specific ECM components or enzymes. For instance, matrix metalloproteinases (MMPs), which are enzymes responsible for the degradation of ECM proteins, have been shown to be upregulated in HD (Silvestroni et al., 2009). Inhibiting the activity of MMPs could potentially help to restore the balance of ECM components and improve neuronal survival and function.

Chemokines, a family of small cytokines, have also been implicated in the pathogenesis of HD due to their role in regulating neuroinflammation and microglial activation (Rocha et al., 2016). Studies have shown that the expression of certain chemokines, such as CCL2 and CXCL12, is altered in the brains of HD patients and animal models, suggesting their potential involvement in the disease progression (Silvestroni et al., 2009). Modulating the activity of chemokines and their receptors has emerged as a promising therapeutic strategy for HD, with preclinical studies demonstrating that the inhibition of CCR2, the receptor for CCL2, can reduce neuroinflammation and improve motor function in HD mouse models (Kwan et al., 2012).

B. Overview of Existing Strategies

Several innovative strategies have been developed to target the mHTT protein at various stages of its expression and function. At the DNA level, genome-editing techniques such as CRISPR-Cas9 have shown promise in selectively targeting and eliminating the expanded CAG repeat in the HTT gene (Yang et al., 2017). This method could potentially correct the genetic mutation and prevent the production of mHTT altogether, as it involves using a guide RNA (gRNA) to direct the Cas9 nuclease to a specific genomic location, enabling targeted editing of DNA by inducing double-strand breaks at that site. Another gene therapy approach involves the use of viral vectors, such as adeno-associated viruses (AAVs), to deliver therapeutic genes that can silence or reduce the expression of mHTT (Miniarikova et al., 2016). AAVs are recognized for their efficacy in gene delivery, capable of transducing both dividing and

non-dividing cells and facilitating sustained expression without genomic integration. However, they face limitations including a restricted capacity for genetic payload and potential immune responses that may hinder repeated administrations.

Figure 3. CRISPR-based Gene Editing Strategy for HD. In a normal chromosome (top), the HTT gene contains a normal CAG repeat length, resulting in the production of wild-type HTT protein. The mutant chromosome (bottom) in HD patients harbors an expanded CAG repeat, leading to the expression of mHTT protein. The CRISPR-based gene editing approach utilizes a guide RNA ($gRNA_1$) that targets the promoter region of the mutant HTT gene, along with a Cas9 endonuclease (not shown) to induce a DSB. The preferred protospacer adjacent motif (PAM) sequence is located near the DSB site. The DSB is then repaired by non-homologous end joining (NHEJ), resulting in small insertions or deletions that disrupt the promoter function and prevent the expression of the mutant HTT gene. A second guide RNA ($gRNA_2$) is used to target the expanded CAG repeat region, leading to its excision and the production of a truncated, non-pathogenic HTT protein. This CRISPR-based gene editing strategy aims to selectively silence the mutant HTT gene while preserving the expression of the normal HTT gene, thereby alleviating the pathogenesis of HD. Adapted from Alkanli et al. (2023).

At the RNA level, ASOs, RNA interference (RNAi), and other RNA-targeting strategies have been explored to reduce the translation of mHTT mRNA into the toxic protein. ASOs are short, synthetic nucleotide sequences that bind to specific regions of the mHTT mRNA and promote its degradation or block its translation (Kordasiewicz et al., 2012). RNAi techniques, including small interfering RNAs (siRNAs), short hairpin RNAs (shRNAs), and microRNAs (miRNAs), exploit the cell's natural RNA silencing pathways to selectively degrade mHTT mRNA (Keiser et al., 2016). siRNAs and shRNAs typically induce mRNA degradation through exact sequence matching, while miRNAs can inhibit translation or cause mRNA degradation despite only partial sequence complementarity. These RNA molecules enable targeted gene silencing, offering a versatile approach to study and manipulate gene expression in various therapeutic applications. However, the efficacy of these RNA tools can be influenced by their stability in the cellular environment and the efficiency of their delivery into target cells.

At the protein level, another promising approach to targeting the mutant huntingtin (mHTT) protein is through the use of monoclonal antibodies (mAbs) directed against the extracellular mHTT. Therapeutic proteins, such as mAbs produced through hybridoma technology, target abnormal proteins with high specificity and are crucial in treating various diseases. Hybridoma technology involves the fusion of

B-cells with immortalized myeloma cells to create a cell line capable of secreting large quantities of specific antibodies. By designing mAbs that selectively bind to and neutralize extracellular mHTT, it may be possible to prevent the spread of the mutant protein between cells and reduce its toxic effects on neighboring neurons (Southwell et al., 2015). This strategy could potentially complement other approaches aimed at reducing the intracellular production or aggregation of mHTT, providing a multi-faceted approach to treating HD. Despite the various therapeutic strategies targeting mHTT at the DNA, RNA, and signaling pathway levels, therapies directly targeting mHTT at the protein level have not been as extensively explored as observed in Figure 4 below, presenting a potential avenue for the development of novel and effective treatments for HD.

Figure 4. Overview of Therapies Associated with mHTT Modification and Degradation. Current therapeutic approaches for HD primarily focus on targeting mHTT at the DNA and RNA levels. Other therapies associated with signaling pathways involve the use of small molecules to modulate the activity of kinases, or to activate autophagy and proteasomal degradation pathways. Stem cell therapy using human induced pluripotent stem cells (iPSCs) is also being explored as a potential treatment option for HD. However, therapies directly targeting mHTT at the protein level have not been as extensively investigated compared to other approaches depicted in this figure. Adapted from Alkanli et al., (2023).

C. Future Potential

These strategies aim to reduce the production and accumulation of mHTT. As stated earlier, one promising approach that builds upon these concepts is the use of AAV vectors to deliver gene therapies targeting mHTT. The paper *"AAV5-miHTT Gene Therapy Demonstrates Sustained Huntingtin Lowering and Functional Improvement in Huntington Disease Mouse Models"* by Spronck et al. (2019) explores this strategy in detail, showcasing the potential of AAV-mediated gene therapy for the treatment of HD. The study's findings are highly relevant to the field of cellular engineering, as they demonstrate the effectiveness of using engineered viral vectors to deliver therapeutic payloads to specific cell types in the brain. In the following section, I will delve deeper into the specifics of the AAV5-miHTT gene therapy study, examining the methods used, the results obtained, and the implications of these findings for the future of HD treatment and cellular engineering in general.

Evaluation of AAV5-miHTT Gene Therapy in HD Mouse Models

I. Objectives and Outcomes

The study demonstrates the potential of AAV5-miHTT gene therapy as a promising approach for the treatment of HD. The authors previously showed that strong HTT lowering using AAV5-miHTT prevented neuronal dysfunction in HD rodents and minipigs after a single intracranial injection. In this study, they further investigated the long-term effects of AAV5-miHTT treatment on HTT protein lowering, tolerability, and functional improvement in two HD mouse models: Q175 knockin (KI) mice and R6/2 mice. Overall, the paper aimed to investigate the long-term efficacy, safety, and tolerability of AAV5-miHTT gene therapy in HD mouse models (Spronck et al., 2019)

II. Creation of Therapy

The AAV5-miHTT gene therapy was created by inserting a gene cassette encoding an engineered microRNA targeting human HTT into an adeno-associated virus serotype 5 (AAV5) vector, which was produced using a baculovirus-based AAV production system and designed to express the therapeutic miRNA under the control of a strong promoter (Spronck et al., 2019). The transcription unit was flanked by two non-coding AAV-derived inverted terminal repeats, creating the AAV vector therapy. AAV5-miHTT was injected into the striatum of Q175 KI mice, and the mice were sacrificed 12 months post-injection. The treatment resulted in a dose-dependent and sustained HTT protein reduction, with subsequent suppression of mutant HTT aggregate formation in the striatum and cortex shown in Figure 5 (Spronck et al., 2019).

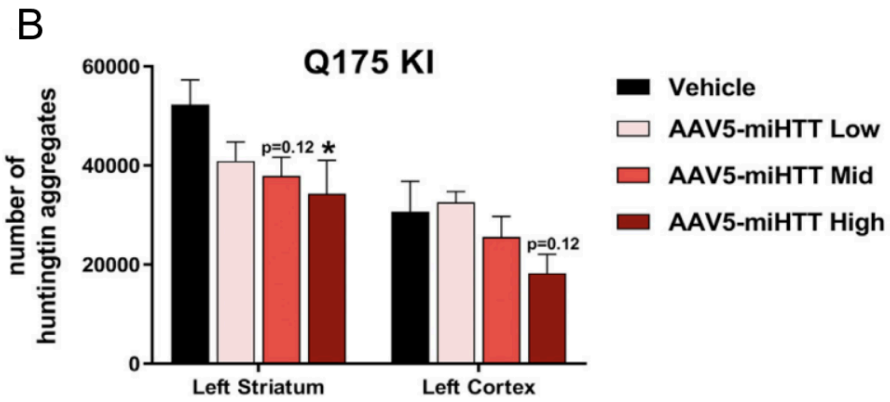


Figure 5. Dose-dependent Reduction of mHTT Aggregates in the Striatum and Cortex of Q175 KI Mice Treated with AAV5-miHTT. (A) Representative images of mHTT aggregates (dark spots) in the striatum and cortex of vehicle-treated and AAV5-miHTT-treated mice at low, medium, and high doses. Reduction can easily be observed. (B) Quantification of mHTT aggregate numbers in the left striatum and left cortex of Q175 KI mice. Data are presented as mean ± SEM. Adapted from Spronck et al. (2019).

III. Proof of Concept

Functional proof of concept was demonstrated in transgenic R6/2 HD mice. Eight weeks after AAV5-miHTT treatment, a significant improvement in motor coordination on the rotarod was observed,

with treated mice showing a 56% increase in latency to fall compared to vehicle-treated R6/2 mice. Additionally, survival analysis showed that a single AAV5-miHTT treatment resulted in a significant 4-week increase in median survival compared to vehicle-treated R6/2 HD mice. The study also included behavioral analyses to evaluate the effects of AAV5-miHTT treatment on HD-like symptoms. Q175 mice underwent rotarod, open field, and fine motor analysis every 3 months after dosing. R6/2 mice were subjected to rotarod testing and clasping behavior assessment (Spronck et al, 2019). The results showed a dose-dependent trend of improvement in motor coordination and clasping behavior in AAV5-miHTT treated animals compared to vehicle-treated controls.

IV. Major Findings and Their Implications

The combination of long-term HTT lowering, reduction in aggregation, prevention of neuronal dysfunction, alleviation of HD-like symptoms, and beneficial survival observed in HD rodents treated with AAV5-miHTT supports the continued development of HTT-lowering gene therapies for HD. The authors conclude that the extensive human mutant HTT lowering, functional improvement, survival benefit, and tolerability of AAV5-miHTT support further development of this gene therapy approach and initiation of clinical trials in HD patients in the near future. Additionally, the study's findings highlight the potential of cellular engineering approaches in the development of novel therapies for neurodegenerative disorders like HD. The one-time bilateral intrastriatal injection of AAV5-miHTT resulted in significant improvements in survival, body weight, motor coordination, and behavior in R6/2 HD mice, partially correcting the severe HD-like phenotype. The median survival was significantly improved by 29 days in the high-dose group compared to untreated R6/2 mice, demonstrating the efficacy of this cellular engineering approach as seen in Figure 6. In summary, the paper provides compelling evidence for the efficacy and safety of AAV5-miHTT gene therapy in HD mouse models. The results demonstrate the ability of this approach to achieve sustained HTT lowering, reduce mutant HTT aggregation, improve motor function, extend survival, and create a favorable safety profile making it a promising candidate for clinical translation coupled with further extensions as discussed later.

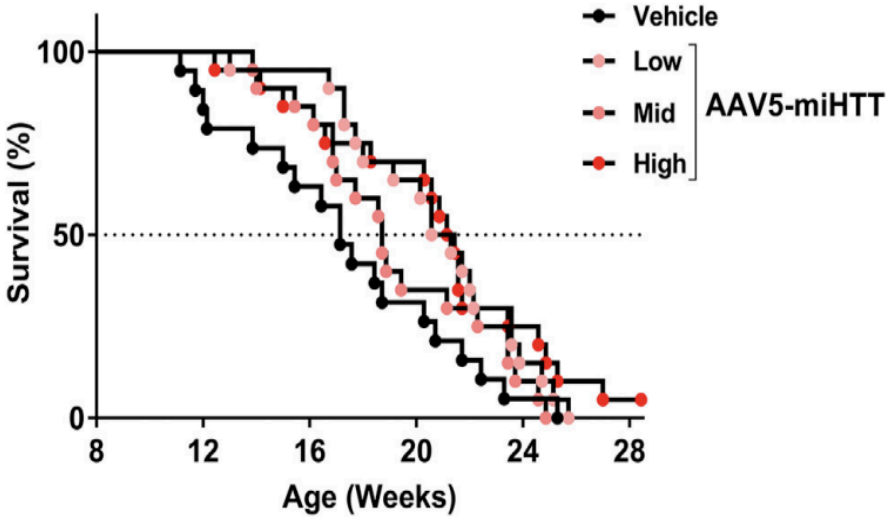


Figure 6. Dose-dependent Effect of AAV5-miHTT Treatment on the Survival of R6/2 Mice. Kaplan-Meier survival analysis demonstrates a significant increase in median survival for R6/2 mice treated with low (26.5 days) and high (29 days) doses of AAV5-miHTT compared to vehicle-treated controls ($p < 0.05$). The survival benefit was more pronounced in the high-dose group, highlighting the potential of AAV5-miHTT gene therapy to extend the lifespan of HD model mice in a dose-dependent manner. Adapted from Spronck et al. (2019).

V. Ties to Cellular Engineering

The authors employed various cellular engineering techniques, including the use of AAV vectors, miRNAs, and genome editing, to create and deliver a therapeutic construct targeting the mHTT gene. One of the key aspects of cellular engineering in this study was the design and production of the AAV5-miHTT vector. As mentioned earlier, the authors used a baculovirus-based AAV production system to create an AAV5 vector encoding a microRNA targeting human HTT (miHTT). The expression of the miHTT cassette was driven by a combination of the cytomegalovirus early enhancer element and chicken β-actin promoter, and the transcription unit was flanked by two non-coding AAV-derived inverted terminal repeats. This design allowed for the efficient delivery and expression of the therapeutic miRNA in the target cells. The study also employed various laboratory techniques commonly used in cellular engineering, such as quantitative PCR (qPCR) and bicinchoninic acid assay (BCA). qPCR was used to determine the amount of vector DNA in the cortex of Q175 KI mice, demonstrating the widespread distribution of the AAV5-miHTT vector through axonal transport in the brain. The BCA assay was used to determine the total protein concentration in brain samples. The use of AAV vectors in this study is particularly relevant to cellular engineering, as AAVs have emerged as promising vehicles for gene delivery in various gene therapy applications. The authors' choice of AAV serotype 5 (AAV5) was based on its ability to effectively transduce the brain, making it an attractive candidate for RNAi-based gene transfer. Furthermore, the authors compared their results with other cellular engineering techniques, such as intracerebroventricular infusions of ASOs targeting HTT mRNA; while the ASO approach showed a 23-day improvement in median survival after daily treatments with a high total dose, the AAV5-miHTT treatment achieved a 29-day improvement with a single intrastriatal injection (Spronck et al, 2019). This difference highlights the potential of AAV-based gene therapy to provide sustained therapeutic benefits with a less invasive and more convenient administration regimen, emphasizing the importance of cellular engineering in optimizing treatment strategies.

VI. Limitations

There are several limitations to the approach given in the paper that should be considered. One of the main limitations is actually the use of AAVs as the main delivery vehicle for the therapeutic miRNA. Although AAVs are widely used in gene therapy applications due to their low immunogenicity and ability to transduce non-dividing cells, they have a relatively low packaging capacity compared to other viral vectors, such as lentiviruses for example. This limits the size of the genetic cargo that can be delivered, potentially restricting the complexity of the therapeutic construct. Additionally, AAV production can be tedious and expensive, which may hinder the scalability and accessibility of the therapy. The cost of producing AAV vectors for clinical trials can range from $10,000 to $100,000 per patient dose, depending on the scale of production (Ferreira et al., 2014). Another limitation of using AAVs is that they do not efficiently integrate the delivered genetic material into the host cell genome, resulting in transient expression of the therapeutic gene. The transient nature of AAV-mediated gene expression may necessitate repeated administrations of the therapy to maintain its efficacy over time, which could increase the risk of adverse effects and reduce patient compliance.

The study also faced challenges related to the targeting of the therapeutic construct to the desired brain regions. Although the intrastriatal injection of AAV5-miHTT aimed to target the striatum, which is primarily affected in HD, dose-dependent levels of vector DNA were observed in the cortex of treated animals. This off-target distribution of the vector could potentially lead to unintended effects on other brain regions and may limit the specificity of the therapy. Another limitation of the study was the insufficient availability of striatum tissue for analysis due to the limited size of the mouse striatum. This restricted the authors' ability to comprehensively assess the effects of the therapy on the primary target region. Furthermore, the fine motor skills and gait analyses performed using the MotoRater failed to detect clear differences between Q175 KI mice and wild-type littermates, preventing the evaluation of the

treatment's impact on these functional parameters. The authors also acknowledged that the observed HTT protein lowering in the CNS was not as strong as expected, possibly due to differences in the assays used and the use of entire brain structures for analysis, which could dilute the effect. Lastly, the choice of the heterozygous Q175 KI mouse model posed limitations to the study. Although this model contains an expanded CAG repeat within one of the native mouse HTT genes, mimicking the mutation found in HD patients, it does not exhibit a robust HD-like phenotype. The lack of clear motor function defects and behavioral deficits in this model prevented the assessment of phenotypic improvement following AAV5-miHTT treatment.

VII. Advancing Human Health

Currently, treatment for HD is limited to managing symptoms and providing supportive care. The development of effective therapies that target the root cause of the disease, such as lowering the levels of mutant HTT protein, is a critical unmet need in the field. The study addresses this need by demonstrating the potential of AAV5-miHTT gene therapy to reduce mutant HTT levels and improve disease phenotypes in HD mouse models.

The authors showed that a single intrastriatal injection of AAV5-miHTT resulted in widespread distribution of the therapeutic miRNA throughout the brain, leading to long-term expression and dose-dependent knockdown of mutant HTT protein in both the striatum and cortex. This reduction in mutant HTT levels was accompanied by a decrease in the number of HTT aggregates, which are a hallmark of the disease. Importantly, despite the limitations discussed, the treatment was well-tolerated, with no adverse effects or histopathological changes observed in any of the major organs, including the brain, up to one year after administration (Spronck et al., 2019).

The relevance of this study to advancing the development of HD therapies is further supported by the authors' previous work in transgenic HD minipigs, which demonstrated the scalability and translatability of the AAV5-miHTT approach to larger, more complex brains. The minipig study showed widespread vector distribution, sustained miHTT expression, and up to 85% reduction in mutant HTT levels, without any tolerability issues. Overall, the study represents a significant step forward in the development of a disease-modifying therapy for HD, demonstrating the potential of AAV5-miHTT gene therapy to reduce mutant HTT levels, improve disease phenotypes, and extend survival in preclinical models. While further research is needed to address the limitations of the current approach and translate these findings to the clinic, this study provides a strong foundation for the continued development of gene therapies targeting mHTT as a promising strategy.

Proposed Treatment Design: Engineered Ubiquitin-Proteasome System Enhancer (UPSE) for Targeted Degradation of mHTT Delivered by a Functional Exosome

I. Background & Motivation

The limitations of the AAV5-miHTT gene therapy approach highlight the need for alternative strategies to combat HD. One promising avenue is to target the mutant huntingtin (mHTT) protein directly at the protein level, rather than at the DNA or RNA level. By focusing on the degradation of the mHTT protein, I can potentially overcome the limitations associated with gene therapy approaches and provide a more targeted and efficient treatment for HD.

II. Proposed Design

The proposed treatment design involves the development of an engineered Ubiquitin-Proteasome System Enhancer (UPSE) that specifically targets mutant huntingtin (mHTT) for degradation. The strategy involves designing a chimeric protein that integrates multiple functional domains to selectively bind mHTT and enhance its degradation via the cell's natural protein degradation machinery—the ubiquitin-proteasome system (UPS)—while preserving the function of the normal huntingtin protein. In HD, the UPS is impaired, leading to the accumulation of mHTT and the formation of toxic protein aggregates. The UPSE aims to restore the normal function of the UPS and promote the efficient clearance of mHTT, thereby alleviating the pathological features of HD. The stability of many proteins is determined by whether they become ubiquitinated, as polyubiquitinated proteins are degraded by a protease complex called the proteasome.

The engineered UPSE will consist of three key components:

1. mHTT-Targeting Nanobody (mHTT-TN): This domain is based on a single-domain antibody fragment, known as a nanobody, which is derived from the variable region of heavy chain antibodies found in camelids. Nanobodies are small, stable, and highly specific, making them ideal for targeting proteins with high affinity (Muyldermans, 2013). The mHTT-TN is generated through a combination of phage display library screening and in vitro affinity maturation techniques to select for nanobodies with exceptional specificity and binding strength for mHTT. The use of nanobodies offers several advantages including improved tissue penetration, reduced immunogenicity, and enhanced stability. Targeting for degradation through the use of a nanobody can be observed in Figure 7.

Figure 7. Schematic Representation of the Nanobody-based Targeting and Ubiquitination of mHTT protein in the UPSE Treatment Design. The mHTT-TN is engineered to specifically bind to the mHTT protein, forming a complex that is recognized by the E3 ubiquitin ligase (E3) component of the UPSE. The E3 ligase, in conjunction with the E1 activating enzyme and E2 conjugating enzyme, facilitates the transfer of ubiquitin molecules to the mHTT protein, resulting in its polyubiquitination. The polyubiquitinated mHTT is then targeted for degradation by the proteasome, effectively reducing the levels of the pathogenic protein in the cell. The incorporation of the mHTT-TN in the UPSE design

enables the specific targeting and elimination of mHTT while preserving the normal functions of the wild-type huntingtin protein. Adapted from Shan et al. (2018).

2. Bifunctional Ubiquitin-Proteasome Adapter (BUPA): This innovative domain combines the functions of a ubiquitin ligase and a proteasome adapter to facilitate the efficient ubiquitination and degradation of mHTT. The BUPA incorporates the Trim-Away technique by utilizing the TRIM21 protein, an E3 ubiquitin ligase that binds to antibody-bound proteins and targets them for proteasomal degradation (Clift et al., 2017). The mHTT-TN nanobody component binds specifically to mHTT, while the TRIM21 component of the BUPA facilitates the rapid ubiquitination of the nanobody-bound mHTT. This combination of the Trim-Away mechanism and the design of the BUPA enhances the efficiency of mHTT degradation by ensuring the fast ubiquitination and proteasomal targeting of the mHTT-nanobody complex.

3. Proteasome Activity Modulator (PAM): To address the impairment of the UPS system in HD, the UPSE includes a domain that can allosterically modulate the activity of the proteasome to enhance its degradation capacity. The PAM is based on a rationally designed peptide that mimics the HbYX motif found in the C-termini of some proteasome activators, such as PA28α and PA200; this motif has been shown to stimulate the peptidase activity of the 20S proteasome by inducing conformational changes in the α-ring subunits (Chu-Ping et al., 1994). Essentially, PAM is an engineered allosteric protein switch that builds synthetic feedback loops modulating the activity of the proteasome generating pathways that display diverse signaling dynamics.

The three components of the UPSE are connected by flexible linker sequences to allow for optimal orientation and functionality of each domain. These linker sequences will be carefully engineered to provide sufficient flexibility and mobility for the individual domains to adopt their active conformations, while also maintaining the overall structural integrity of the entire protein complex (Chen et al., 2013).

III. Delivery

The use of exosomes as a nanocarrier system for the delivery of the UPSE to the brain represents a cutting-edge approach to overcome the challenges associated with crossing the blood-brain barrier (BBB). Exosomes are small, natural, lipid-bilayer vesicles secreted by various cell types that play a crucial role in intercellular communication. They have unique properties such as low immunogenicity, high stability, and ability to cross biological barriers, including the BBB (Batrakova & Kim, 2015). A diagram showcasing exosomes and their potential can be seen in Figure 8.

To create a functionalized exosome capable of delivering the UPSE specifically to the affected brain regions, I will engineer the exosomes to display specific surface ligands that can facilitate their crossing of the BBB and enhance their uptake by neurons. One promising approach is to incorporate the rabies virus glycoprotein (RVG) peptide onto the surface of the exosomes; the RVG peptide has been shown to bind specifically to the acetylcholine receptor (AchR) expressed on the surface of neuronal cells and the brain capillary endothelial cells that form the BBB (Kumar et al., 2017). By displaying the RVG peptide, the functionalized exosomes can effectively cross the BBB through receptor-mediated transcytosis and be selectively taken up by the neuronal cells in the affected brain regions.

To produce the functionalized exosomes, I will genetically engineer a suitable cell line, such as mesenchymal stem cells (MSCs), to express the UPSE along with the RVG peptide on the exosomal surface. The engineered MSCs will be cultured in a bioreactor system under optimized conditions to promote the secretion of UPSE-loaded, RVG-displaying exosomes. The exosomes will be harvested from

the cell culture medium using advanced purification techniques, such as size-exclusion chromatography and affinity-based isolation, to ensure a high purity and homogeneity of the final product (Sidhom et al., 2020). The purified functionalized exosomes containing the UPSE will be formulated into an injection that can be administered to patients via IV. Once injected into the bloodstream, the exosomes will navigate through the circulatory system and cross the BBB, thanks to the RVG peptide displayed on their surface. Upon reaching the brain, the exosomes will be internalized by the neuronal cells via AchR-mediated endocytosis, releasing their UPSE cargo into the cytoplasm. The UPSE will then bind to mHTT, facilitating its ubiquitination and degradation by the proteasome, as described earlier.

The use of functionalized exosomes as a nanocarrier system for the targeted delivery of the UPSE offers several advantages over other delivery methods. First, exosomes are natural, biocompatible, and non-immunogenic, reducing the risk of adverse immune reactions. Second, the ability to engineer the exosomal surface with specific ligands, such as the RVG peptide, enables the selective targeting of the affected brain regions and enhances the efficiency of drug delivery. Third, the small size and unique composition of exosomes allow them to cross the BBB more effectively compared to other nanoparticle-based delivery systems. Finally, the use of a non-invasive, systemic delivery method, such as intravenous injection, makes the treatment more accessible and convenient for patients, eliminating the need for invasive surgical procedures. Commercial product development for exosomes is still in infancy, and thus its application should be cautious but the research area is wide open.

Figure 8. Schematic Representation of Exosome Biogenesis, Secretion, and Uptake by Recipient Cells. Exosomes are formed by the inward budding of the endosomal membrane, creating intraluminal vesicles (ILVs) within multivesicular bodies (MVBs) as depicted to the left. MVBs can either fuse with lysosomes for degradation or with the plasma membrane, releasing the exosomes into the extracellular space. Exosomes contain various biomolecules, such as proteins, lipids, DNA, and RNA, which can be transferred to recipient cells through endocytosis or receptor-mediated cell signaling, facilitating intercellular communication. Adapted from Huo et al., 2021.

IV. Development, Testing, and Advantages

The engineered UPSE will be expressed in a suitable mammalian cell line and purified using affinity chromatography. The purified protein will be characterized in vitro to assess its binding specificity, ubiquitination efficiency, and UPS activation capabilities. The efficacy of the UPSE in promoting mHTT degradation will be evaluated in a series of in vitro and in vivo studies.

In vitro experiments include binding assays to assess the specificity and affinity of the mHTT-TN, ubiquitination assays to evaluate the efficiency of the BUPA, and proteasome activity assays to determine

the effectiveness of the PAM. Cell culture models of HD, such as patient-derived induced pluripotent stem cell (iPSC)-derived neurons, using techniques like Western blot analysis and immunofluorescence imaging, will be conducted to assess the ability of the UPSE to reduce mHTT levels and alleviate cellular pathology.

To test the therapeutic potential of the engineered UPSE in vivo, I will employ a newfound representative zQ175 knock-in mouse model, which exhibits a robust HD-like phenotype; this novel humanized HD mouse model expresses the full-length human mHTT gene under the control of its endogenous promoter (Menalled et al., 2012). I will administer the engineered protein to zQ175 mice via intravenous injection of functionalized exosomes carrying the UPSE. The treated mice will undergo a series of behavioral tests, including the rotarod test, open field test, and grip strength test, to assess motor function, coordination, and overall physical performance at various time points post-treatment. Additionally, I will collect brain tissue samples from the treated mice and analyze them using biochemical and histological techniques, such as western blotting, immunohistochemistry, and qPCR, to quantify the levels of mHTT, examine the extent of neuronal degeneration, and evaluate the expression of key markers of HD pathology.

To assess the safety and long-term efficacy of the UPSE, a comprehensive preclinical development program will also be implemented. This includes pharmacokinetic and pharmacodynamic studies to evaluate the absorption, distribution, metabolism, and excretion of the UPSE. Immunogenicity studies will also be conducted to evaluate the potential immune response to the UPSE and to develop strategies for mitigating any adverse immune reactions; for example, I could administer UPSE to a mammalian model, then measuring antibody levels and T-cell responses over time. In addition to the preclinical studies, the development of the UPSE is supported by a robust manufacturing and quality control process. This includes the establishment of a scalable and reproducible manufacturing protocol along with the implementation of quality control measures to ensure the consistency and safety of the final product.

The proposed treatment design offers several advantages over the AAV5-miHTT gene therapy approach. First, the UPSE targets mHTT at the protein level, rather than at the DNA or RNA level, allowing for a more direct and efficient clearance of the disease-causing protein as it bypasses the need for transcriptional or translational regulation. Second, the use of nanobodies as the mHTT-targeting domain provides improved specificity, tissue penetration, and reduced immunogenicity compared to traditional protein binder domains. Third, the incorporation of the BUPA and PAM components enhances the efficiency of mHTT ubiquitination and degradation, while also restoring the normal function of the UPS. This targeted delivery approach addresses the limitations of the AAV5-miHTT gene therapy study, which showed off-target distribution of the vector in the cortex despite intrastriatal injection. Furthermore, the modular design of the UPSE allows for the optimization of its individual components to enhance its overall performance. For example, the mHTT-TN can be fine-tuned to improve its specificity and affinity for mHTT, while the BUPA and PAM can be optimized to increase the efficiency of mHTT ubiquitination and proteasomal degradation. This modularity provides opportunities for further refinement and adaptation of the treatment design based on experimental results.

V. Pioneering UPSE Therapeutics for the Future

The use of advanced protein engineering and drug delivery technologies, combined with a comprehensive preclinical development program, positions the UPSE as a promising candidate for clinical translation. This would also serve as a proof-of-concept for the development of targeted protein degradation therapies for other neurodegenerative disorders, such as Alzheimer's disease, Parkinson's disease, and amyotrophic lateral sclerosis. The principles and technologies underlying the UPSE, such as the use of nanobodies, bifunctional adapters, and proteasome activity modulators, could be adapted to target a wide range of

disease-causing proteins, opening up new possibilities for the treatment of previously intractable neurodegenerative conditions. The modular nature of the engineered protein allows for optimization and adaptation based on experimental results, while the combination of in vitro and in vivo studies will provide a comprehensive assessment of its therapeutic potential. The UPSE offers a promising avenue for the development of disease-modifying therapies that could transform the lives of patients worldwide. Ultimately, the successful development and implementation of this novel treatment design could lead to a significant advancement in the field of HD therapeutics, offering hope for patients and families.

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